

SOUTH ARKANSAS, AR, USA

WMO: 723419

Lat: 33.221N

Lon: 92.814W

Elev: 252

StdP: 14.56

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93992

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.0	25.3	9.0	8.8	28.8	14.0	11.3	32.1	19.2	59.9	17.9	57.8	3.6	20	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	98.6	76.7	95.7	76.6	93.5	76.7	80.3	90.5	79.3	90.0	78.5	88.7	6.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.7	145.1	83.4	76.8	141.0	83.0	75.8	136.0	82.5	43.8	90.7	42.9	89.9	42.1	89.0	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.3	15.2	12.8	15.6	102.1	4.2	4.9	12.6	105.6	10.1	108.4	7.8	111.1	4.7	114.7		
			14.1	82.4	4.0	2.2	11.2	84.1	8.9	85.4	6.6	86.6	3.7	88.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	44.8	48.3	56.0	63.9	71.9	79.0	82.0	81.9	75.7	64.5	53.9	46.7
	DBStd	15.47	10.37	10.27	9.44	7.51	6.16	4.11	3.76	4.31	6.05	8.10	8.90	9.78
	HDD50	676	231	142	49	4	0	0	0	0	0	3	60	187
	HDD65	2567	631	472	303	108	18	0	0	0	5	111	346	573
	CDD50	5835	69	96	235	421	679	870	993	988	770	452	178	84
	CDD65	2252	4	6	24	75	233	420	528	523	325	95	14	5
	CDH74	22945	8	35	211	699	2075	4253	5764	5761	3122	916	86	15
	CDH80	10706	0	2	30	135	713	2009	2978	3088	1458	286	7	0
Wind	WSAvg	5.4	6.1	6.4	6.7	6.8	5.7	4.8	4.3	4.2	4.3	4.6	5.2	5.7
Precipitation	PrecAvg	48.1	4.7	3.7	5.0	4.8	5.5	3.1	4.4	3.3	3.8	3.2	3.6	4.3
	PrecMax	73.5	9.6	6.7	9.4	9.8	13.1	6.6	9.3	7.9	8.5	9.0	9.5	9.3
	PrecMin	35.1	1.4	0.8	1.7	0.5	0.5	0.2	0.6	0.2	0.5	0.0	1.2	0.6
	PrecStd	12.5	2.5	1.7	2.2	3.0	3.1	2.1	2.2	2.1	2.4	2.6	2.2	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.9	78.7	83.5	86.6	91.2	97.4	100.4	102.8	98.6	90.1	81.1	76.2
		MCWB	63.9	65.0	65.7	70.6	75.1	76.0	76.9	75.9	74.5	71.1	68.4	66.6
	2%	DB	71.0	73.5	79.4	83.6	89.1	94.4	97.4	99.4	94.6	86.4	76.3	71.7
		MCWB	62.0	62.9	64.9	68.2	73.8	76.4	77.4	76.7	74.4	70.9	65.5	64.5
	5%	DB	67.0	70.0	75.7	81.2	86.8	92.0	95.0	96.3	91.4	83.3	72.9	67.8
		MCWB	60.6	61.7	62.8	67.2	73.0	76.2	77.6	76.6	74.2	68.7	64.2	62.8
	10%	DB	63.2	66.1	72.1	78.4	84.3	90.1	92.9	93.5	88.6	79.5	69.6	63.8
		MCWB	57.5	58.8	61.2	65.6	72.0	75.9	77.5	76.8	73.5	67.4	63.4	58.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.0	68.6	69.4	74.2	78.5	80.4	81.8	81.4	79.0	76.9	71.9	68.8
		MCDB	72.4	73.6	76.9	81.9	87.5	90.2	90.9	91.0	88.9	84.8	75.5	72.9
	2%	WB	64.8	66.2	67.8	71.6	76.4	79.1	80.3	80.2	77.8	74.7	69.0	66.5
		MCDB	69.1	71.5	75.6	79.0	84.8	89.0	90.6	90.6	87.9	81.8	73.3	70.3
	5%	WB	62.4	63.8	65.8	70.3	75.1	78.2	79.5	79.1	76.8	72.7	66.9	63.8
		MCDB	65.9	68.2	72.6	77.3	83.2	87.9	90.4	89.9	86.4	78.4	71.2	67.2
	10%	WB	58.3	60.4	63.6	68.6	73.8	77.3	78.6	78.2	75.6	70.6	64.0	60.0
		MCDB	62.3	65.0	70.0	75.3	81.5	86.7	89.0	88.9	84.5	76.0	68.3	63.2
Mean Daily Temperature Range	5% DB	MDBR	20.8	21.2	22.7	23.3	20.9	20.4	20.3	21.7	22.8	24.0	22.3	20.3
		MCDBR	24.4	25.4	25.7	24.5	22.7	22.8	23.4	25.9	25.9	27.3	25.6	23.2
		MCWBR	16.9	16.2	13.1	11.4	9.4	8.0	7.4	7.4	8.8	12.7	15.8	16.2
	5% WB	MCDBR	19.3	21.1	21.5	19.7	19.2	19.8	20.6	21.3	21.0	20.2	20.3	19.7
		MCWBR	15.8	15.8	13.4	11.3	9.0	8.0	7.8	7.7	8.0	11.7	15.4	15.5
Clear-Sky Solar Irradiance	taub	0.327	0.337	0.364	0.395	0.435	0.456	0.483	0.476	0.419	0.363	0.345	0.326	
	taud	2.514	2.484	2.420	2.327	2.274	2.259	2.208	2.222	2.357	2.484	2.488	2.522	
	Ebn at Noon	282	291	290	284	272	265	257	257	268	277	273	276	
	Edh at Noon	27	30	35	40	43	43	46	44	37	30	27	25	
All-Sky Solar Radiation	RadAvg	829	945	1344	1674	1849	2016	1974	1839	1576	1268	911	726	
	RadStd	80	158	115	112	141	157	129	155	142	173	104	87	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page